## PEER REVIEW HISTORY

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## ARTICLE DETAILS

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| **TITLE (PROVISIONAL)** | The effectiveness of rehabilitation interventions on the secondary consequences of surviving a cardiac arrest: a systematic review and meta-analysis |
| **AUTHORS** | Joshi, Vicky; Christensen, Jan; Lejsgaard, Esben; Taylor, Rod; Zwisler, Ann Dorthe; Tang, Lars |

## VERSION 1 – REVIEW

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| **REVIEWER** | Lilja, Gisela<br>Lund University, Department of Clinical Sciences |
| **REVIEW RETURNED** | 06-Jan-2021 |

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| **GENERAL COMMENTS** | Thank you to the authors for a very relevant paper. During the recent years, the focus on survivorship after CA has increased. Long-term consequences have been described to affect the patients and their relatives even months and years after the success of the initial resuscitation. The effects on participation and quality of life has been described, despite quality of life in general being less affected and in the majority similar to the general population. Although the high prevalence and effects on "secondary consequences", most interventions are still focused on survival, and few interventions are described and recommended after the hospital stay. A systematic review of the current availability of effective interventions to support outcome after the hospital stay is much needed, not least to guide the current knowledge gap and support future studies. We do however have some comments.<br><br>Major comments<br>- The definition of rehabilitation in the manuscript introduction does not match the usual definition of rehabilitation. Is it really rehabilitation interventions you are referring to, or more general non-pharmacological interventions? We suggest that you specify and define this a bit clearer.<br>- We understand that the primary and main secondary outcomes of health-related quality of life (HRQoL) and neurological function is based on the COSCA work – at the same time we are a bit concerned if this is a limitation in terms of rehabilitation. A framework commonly used in rehabilitation is the ICF where the goal mostly is to help the person to be as independent as possible in everyday activities and enable participation in e.g. work. leisure activities and other roles. A successful rehabilitation may be seen by improving functions (such as neurological functions), but does not have to be. A successful rehabilitation may be seen also when no improvements in neurological function is seen, e.g. by finding new ways to perform activities (strategies, aids, environmental changes) participation may still be improved, and emotional well- |

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| | being may become better with support to adjust to the new situation. We would therefore suggest that you describe the in terms of rehabilitation so important ICF, and the outcomes you refer to in relation to that, to help the reader to see the full picture.<br>- All conclusions should include that this result is related mainly to the specific outcome of interest (= HRQoL and not e.g. activity and participation) AND the substantial lack of studies in the area.<br>- The COSCA outcomes are, as the authors state, chosen based on consensus by patients, relatives, researchers, professionals etc. and is definitely important. However, the way we measure the outcome of HRQoL is mostly by generic HRQoL tools that are known to be crude (especially if composite scores are used, see e.g. Bohm et al 2019 and work by Dr. Haywood). Important details about the patients' outcome may therefore be lost, and you will need very large differences to be able to see an effect. when the outcome is too crude. We would like you to include this important bias in the manuscript. To note this limitation is also highlighted in the COSCA paper. The COSCA therefore recommends to add complimentary detailed and condition specific outcomes depending on the aim of the study which is probably very relevant for rehabilitation studies to not loose smaller but still very clinical important effects. A concern in the context of this review is the lack of discussion regarding this.<br>- The term standard care should be discussed. It is always a problem in rehabilitation studies that you cannot deny someone standard care. Standard care varies and may in some cases be absent, but may in other situations be quite ok or even good. It is an important problem that you do not have control of what you measure against and this may influence the results.<br>- Cognitive impairment is one of the most common and important consequences after a cardiac arrest. We therefore miss cognition as an outcome, or at least a discussion of the lack of cognitive rehabilitation interventions. In relation to this we would like you to add more information on how additional secondary outcomes were chosen. Why for example are not cognition (impairment), ADL (activity) or RTW (participation) included?<br>- The outcome measure used may have important effects on the results, and the fact that measures of neurological function included both test of cognition or physical ability, and may have been either observer or patient-reported need further clarification and discussion (Methods: Eligibility criteria). Have also performance-based tests been used in the relevant studies? Have observer-reports consisted of clinician-reports (as we suspect is the case with FIM) only, or also of informant-reports involving next of kin or a close friend? If so, please state that this has been the case. We also know from studies on other acquired brain injuries or neurodegenerative disorders that the overlap between subjective (patient-reported) and objective outcome is not 100%. Furthermore, cut-points and instruments used to describe e.g. cognitive impairment varies between studies. The usage of pooled SF-36 data has been discussed as a limitation for the HRQoL data, but the authors may similarly want to discuss the implications of this as well.<br>- Suggestion: Include more description of what we know about rehabilitation interventions in associated groups of patients (that we know often include a few CA survivors) e.g. brain injury rehabilitation, cognitive rehabilitation, psychological interventions, cardiac rehabilitation etc. to show that there are available interventions that could be tested for this group specifically as well. |

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| | Minor comments<br>Abstract<br>- Secondary consequences in the aim is a bit unclear, please specify what you mean with this.<br>- We are uncertain of the dash usage in "8-weeks" and "2-years".<br>- The SMD of physical but not mental HRQoL is reported in the abstract.<br>Introduction<br>- SCA is primarily used to describe sudden cardiac arrest, even in scientific literature. This abbreviation appears throughout the manuscript and is somewhat confusing. Have you considered another abbreviation for survivors of cardiac arrest (such as CA-survivors)?<br>- Secondary physical, neurological and psychological consequences – physical may also be of neurological reason. Do you actually mean physical, cognitive and psychological?<br>- Page 3 row 11-13. We are not sure about the wording here ("in people").<br>Methods<br>- Summary measures. A suggestion is that the authors here or somewhere else briefly specify the "acceptable" range for SMD and I2, for instance. This may be helpful for readers not familiar with these statistical methods.<br>Results<br>- We would not consider FIM as a measure of neurological function. FIM is related mostly to activity even if it includes some items regarding neurological or cognitive impairment. Problems may however not only be related to neurological function.<br>- We are not sure if p=0.00 should rather be written out as p<0.001 throughout this section. Also, the heading "HRQoL meta-analysis" should be in italics as well.<br>- Please be consistent regarding spacing before/after +/- signs. Double check the number of parentheses used (e.g. in section "Synthesis of results, HRQoL studies not included in meta-analysis").<br>Discussion<br>- The discussion of the different populations in the studies could be expanded. Some studies have only included CA related to cardiac cause, other studies have included patients with anoxic BI due to other reasons as e.g., suicide attempts or drug overdoses. This may influence on the results as well as the mix of IHCA and OHCA.<br>- We do not follow the current statement "from these only associated change in outcomes can be determined not effectiveness" (page 15 row 9-10).<br>- We would classify stroke as an acquired brain injury (page 15 row 50). Consider rephrasing.<br>Table 1<br>- Consider writing out the explanation for ICD as a footnote to the table.<br>Overall<br>- Be consistent in the use of primary outcome, main secondary outcome or secondary outcome. Define what is the primary or secondary outcome and describe this similar all over the manuscript and be clear in conclusions what outcome you refer to (e.g., if it is HRQoL or other). |

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| **REVIEWER** | Bosch, Frank H.<br>Hospital Rijnstate Arnhem, Internal Medicine |
| **REVIEW RETURNED** | 11-Jan-2021 |

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| **GENERAL COMMENTS** | This study is well done research. The results are not surprising, but it is good that it is now established that there is not much literature on this topic; there are a few minor remarks regarding the text.<br>Page 3: i think the authors meant (≥ 18 years)<br>Page 5: continues data (i am not sure what they mean)<br>Page 6: GRADE was used; additional analyses: will be investigated; |

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| **REVIEWER** | Prescuitti, Alex<br>University of Colorado, Psychology |
| **REVIEW RETURNED** | 06-Feb-2021 |

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| **GENERAL COMMENTS** | Thank you for inviting me to review this systematic review and meta-analysis on the effectiveness of rehabilitation interventions on secondary outcomes post-cardiac arrest. This is a timely and very important research question, given the call for multi-disciplinary rehabilitation efforts for cardiac arrest survivors. The authors found low quality evidence from 3 RCTs and 11 observational studies, and thus call for high quality RCTs comparing rehabilitation interventions with standard care alone. The study authors should be credited for utilizing an extensive search strategy and for utilizing appropriate tools to ensure the integrity of their appraisal of individual studies and subsequent analyses.<br><br>Overall, the message is clear – high quality RCTs are needed to evaluate the effectiveness of rehabilitation interventions, and particularly, specific components of these interventions, in cardiac arrest survivors.<br><br>I only have a few points for consideration:<br><br>Results:<br><br>1. Dougherty 2004,2005 does not appear to be a rehab intervention but rather an information packet aimed at increasing self-efficacy and reduce anxiety. I would consider excluding it.<br><br>2. It is unclear if Dougherty 2019 was a rehab intervention based on limited detail provided.<br><br>Minor:<br><br>Methods:<br><br>1. Edit to ensure tense remains consistent in the Methods section as the authors used both past and present throughout.<br><br>2. At points the primary aim of the study is reported to be examining effectiveness of rehab interventions, with safety being a secondary outcome. At other points the primary aim is reported to be examining effectiveness and safety. Please edit for consistency. |

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| **REVIEWER** | Esterman, Adrian<br>University of South Australia, School of Nursing and Midwifery |
| **REVIEW RETURNED** | 18-Apr-2021 |

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| **GENERAL COMMENTS** | Page 4: Introduction, line 47 - Should read ≥18<br>Page 6: Summary measures, line 35- Should read continuous data<br>Page 6: Synthesis of results, line 47 – Should read Study heterogeneity, not Study variance<br>Page 13: Synthesis of results, lines 31-35 – please use p<0.001 for values close to zero.<br>Page 13: Line 41, please do not use ±. If these are standard deviations, say so. This is true throughout the results.<br>Pages 29 and 30: Both forest plots labelled Figure 4B! In the first, I2 was 0, yet a random effects model was still used. Also for Table 5, I2 was 17%, yet a random effects model was still used |

**VERSION 1 – AUTHOR RESPONSE**

Response to reviewers. Systematic revie

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| Reviewer: 1 | | |
| Dr. Gisela Lilja, Lund University | | |
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| Comments to the Author: | | |
| Thank you to the authors for a very relevant paper. During the recent years, the focus on survivorship after CA has increased. Long-term consequences have been described to affect the patients and their relatives even months and years after the success of the initial resuscitation. The effects on participation and quality of life has been described, despite quality of life in general being less affected and in the majority similar to the general population. Although the high prevalence and effects on "secondary consequences", most interventions are still focused on survival, and few interventions are described and recommended after the hospital stay. A systematic review of the current availability of effective interventions to support outcome after the hospital stay is much needed, not least to guide the current knowledge gap and support future studies. We do however have some comments. | | |
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| Major comments | | |
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| - The definition of rehabilitation in the manuscript introduction does not match the usual | We agree that the definition of rehabilitation we have | *Text added to Introduction Page 3 Line 6:* |

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| definition of rehabilitation. Is it really rehabilitation interventions you are referring to, or more general non-pharmacological interventions? We suggest that you specify and define this a bit clearer. | used in the manuscript needs further clarification. No unique definition of rehabilitation can be found in the literature. This typically changes in relation to profession, health setting and patient populations. In designing the review protocol we used the White book definition:<br>'A set of measures that assist individuals, who experience or are likely to experience disability, to achieve and maintain optimum functioning in interaction with their environments.'<br>(From: Cantista P, Ceravolo MG, Christodoulou N, et al. White Book on Physical and Rehabilitation Medicine (PRM) in Europe. Chapter 1. Definitions and concepts of PRM. *European Journal of Physical and Rehabilitation Medicine* 2018;54(2):156-65. doi: 10.23736/S1973-9087.18.05144-4).<br><br>As this is the first review of rehabilitation interventions for CA-survivors we wanted to include all possible rehabilitation interventions. We therefore used the White book definition since as it gives a broad difination of rehabilation in line with the broad needs of CA-survivors and the aim of this review. Hence, the PICO 'intervention' eligibility criteria were defined not be inclusion of specific interventions such as physiotherapy or cardiac rehabilitation but, in alignment with the White book definition of rehabilitation, by the exclusion of primarily pharmacological, surgical or invasive technology.<br><br>We have added a sentence to describe the aim of | "Rehabilitation helps people to achieve and maintain optimum functioning in interaction with their environments.[7]"<br><br>*Text added under Methods, eligibility criteria Line 4, Page 4:*<br><br>Rehabilitation can be defined as: 'A set of measures that assist individuals, who experience or are likely to experience disability, to achieve and maintain optimum functioning in interaction with their environments'.[7] To align with this broad definition of rehabilitation and ensure inclusion of all possible rehabilitation interventions, interventions were included if they were not primarily pharmacologically or surgically-based or involve invasive technology. Interventions in the emergency room or critical care unit setting were excluded<br><br>*Text added to Methods, Searches Line 39, Page 4:*<br>The search matrix consisted of a combination of keywords and synonyms for: 1) cardiac arrest, and 2) non-pharmacological/surgical/invasive technology rehabilitation interventions |

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| | rehabilitation to the introduction and added a definition of rehabilitation to back support our intervention inclusion criteria to the Methods section. We have added detail to the search matrix terms to align with our description of included interventions. | |
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| - We understand that the primary and main secondary outcomes of health-related quality of life (HRQoL) and neurological function is based on the COSCA work – at the same time we are a bit concerned if this is a limitation in terms of rehabilitation. A framework commonly used in rehabilitation is the ICF where the goal mostly is to help the person to be as independent as possible in everyday activities and enable participation in e.g. work. leisure activities and other roles. A successful rehabilitation may be seen by improving functions (such as neurological functions), but does not have to be. A successful rehabilitation may be seen also when no improvements in neurological function is seen, e.g. by finding new ways to perform activities (strategies, aids, environmental changes) participation may still be improved, and emotional well-being may become better with support to adjust to the new situation. We would therefore suggest that you describe the in terms of rehabilitation so important ICF, and the outcomes you refer to in relation to that, to help the reader to see the full picture. | These are some very fine thoughts. We agree that there are many outcomes of relevance in regards to rehabilitation intervention. One could argue that outcomes of interest in this review primarily related to the first part of ICF (body and function) and less on activity and participation.<br><br>In order to contain the size of the review we needed to place some limits on the number of outcome categories that could be described, analyzed and discussed. We believe that we have included the most common outcomes used in rehabilitation, however, we agreed that activity and participation is an objective of rehabilitation and we have suggested that future systematic reviews could focus on the effect of interventions on activity and participation and other outcomes not included in the present study. | *Text added to Discussion, Implications for future research and clinical practice Line 28, Page 18:*<br>"This systematic review has focused primarily on impairment and function outcomes and less on activity and participation. Hence, future systematic reviews on this subject could consider including these outcomes." |
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| - All conclusions should include that this result is related mainly to the specific outcome of interest (= HRQoL and not e.g. activity and participation) AND the substantial lack of studies in the area. | We have clarified that the conclusions of the systematic review relate to the included outcomes. | *Text added to the Abstract, Conclusion Line 30, Page 1 and manuscript Conclusion Line 40, Page 18 :*<br>"Given the overall low quality of evidence, this review cannot determine the effectiveness of rehabilitation interventions for survivors of cardiac arrest on HRQoL, neurological function or |

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| | | other included outcomes, and recommend further high-quality studies be conducted." |
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| - The COSCA outcomes are, as the authors state, chosen based on consensus by patients, relatives, researchers, professionals etc. and is definitely important. However, the way we measure the outcome of HRQoL is mostly by generic HRQoL tools that are known to be crude (especially if composite scores are used, see e.g. Bohm et al 2019 and work by Dr. Haywood). Important details about the patients' outcome may therefore be lost, and you will need very large differences to be able to see an effect. when the outcome is too crude. We would like you to include this important bias in the manuscript. To note this limitation is also highlighted in the COSCA paper. The COSCA therefore recommends to add complimentary detailed and condition specific outcomes depending on the aim of the study which is probably very relevant for rehabilitation studies to not loose smaller but still very clinical important effects. A concern in the context of this review is the lack of discussion regarding this. | Thank you for highlighting the potential limitations of studies using generic versus disease specific HRQoL tool. We have highlighted this in the Discussion and suggested future studies consider including disease specific outcomes in the section on Implications for Future practice. | *Text added to the Discussion Line 39, Page 17:*<br>"Our primary outcome, HRQoL, is an important outcome in rehabilitation research.[7] However, the choice of generic or disease specific HRQoL measures may influence the results as generic measures of HRQoL can be crude with important details lost and large sample sizes required to demonstrate effect[4 45 46]. In this review all studies except one used generic measures of HRQoL."<br><br>*Text added to Implications for future research Line 21, Page 18:*<br>A minimum outcome set for these future rehabilitation RCT should include those recommended by COSCA[4], HRQoL and neurological function and consider including disease specific outcomes. |
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| - The term standard care should be discussed. It is always a problem in rehabilitation studies that you cannot deny someone standard care. Standard care varies and may in some cases be absent, but may in other situations be quite ok or even good. It is an important problem that you do not have control of what you measure against and this may influence the results. | We agree standard care provided to the control groups in RCT can influence the results.<br>We have added text on this to the Discussion | *Text added to Discussion Line 1, Page 18:*<br>"Another element that potentially influenced our findings may be the standard care received by the RCT control groups. Two[21-23] of the included RCT provided educational elements to both the intervention and control groups and in a third[47] participants could have received cardiac rehabilitation." |
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| - Cognitive impairment is one of the most common and important consequences after a cardiac arrest. We therefore miss | We agree that cognitive impairment is a common and important consequence after cardiac | *Text added to Results, neurological function Line 37, Page 13:* |

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| cognition as an outcome, or at least a discussion of the lack of cognitive rehabilitation interventions. | arrest. As we used the core domain descriptors from COSCA, cognition was included under neurological function but we have tried to highlight cognition more clearly in the text.<br><br>We have clarified that the results relate to cognitive tests in the first part of the report of results under neurological function.<br><br>A second trial registration was discovered during the updated search/screen process. We have highlighted that this study is investigating cognitive outcomes, specifically executive function.<br><br>In the discussion we have noted, as you suggest, that there are a lack of rehabilitation intervention with cognition as their main focus. | The RCT[30] showed an outpatient rehabilitation intervention had no significant effectiveness in improving cognitive function on performance-based cognitive tests<br><br>*Text added to Results, neurological function data not included in meta-analysis Line 16, Page 14:*<br>"Cognition, specifically executive function, is the primary outcome in one included ongoing RCT,[35] with results due in 2024."<br><br>*Text added to Discussion Line 40, Page 16:*<br>"This review found very few studies aimed at improving neurological function including cognition for CA-survivors. However, one ongoing RCT was found investigating a computer-based intervention to improve executive function with results due in 2024." |
| In relation to this we would like you to add more information on how additional secondary outcomes were chosen. Why for example are not cognition (impairment), ADL (activity) or RTW (participation) included? | We agree that further clarification of the secondary outcomes should be included. Secondary outcomes were inspired from existing literature on the consequences of cardiac arrest for survivors as well as existing systematic reviews investigating cardiac rehabilitation.<br>We have added this detail to the Methods, eligibility criteria section.<br><br>We agree that there are many outcomes of relevance in regards to rehabilitation interventions. As we previously mentioned, one could argue that outcomes of interest in this review primarily related to the first part of ICF (body and function) and less on activity and participation. These were chosen in order to contain the size of | *Text added to Methods, Eligibility criteria Line 19, Page 4:*<br>"The primary and main secondary outcomes were chosen as, alongside survival, HRQoL and neurological function have been identified as important core outcome domains after cardiac arrest by survivors, relatives and clinicians.[4 13]"<br><br>"Choice of secondary outcomes was informed by existing evidence on the secondary consequences of cardiac arrest[1 3 5 6] and inspired by outcomes in previous systematic reviews on rehabilitation with other cardiac disease populations.[8 15]"<br><br>*Text added to Discussion, implications for future research and clinical practice Line 28, Page 18*:<br>"This systematic review has focused primarily on impairment and function outcomes and less on activity and participation. Hence, future systematic reviews |

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| | the review and limits on the number of outcome categories so that these and our finding could be better described, analyzed and discussed. We still believe that we have included the most common outcomes used in rehabilitation, however, we agreed that activity and participation is an objective of rehabilitation and we have suggested that future systematic reviews could focus on the effect of interventions on activity and participation and other outcomes not included in the present study. | on this subject could consider including these outcomes." |
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| - The outcome measure used may have important effects on the results, and the fact that measures of neurological function included both test of cognition or physical ability, and may have been either observer or patient-reported need further clarification and discussion (Methods: Eligibility criteria). | We agree that outcomes can come from different sources. We have acknowledged this in the Eligibility criteria.<br>And have noted the source of outcome when reporting the results for neurological function.<br><br>We agree that outcomes measures may produce different results depending on whether they are observer or patient reported. In our meta-analysis we did, however, not include outcome measures from different sources and therefore this did not influence the review's results. Hence, given the limited space, we have decided not to include this point in the discussion but have instead focused on some of the other considerations that may have influenced the review's results. We hope the reviewer agree with our priorities. Otherwise please let us know. | *Text added to Eligibility criteria Line 18, Page 4:*<br>"Measures may be patient-reported, clinician-reported, observer-reported or performance-based".<br><br>*Text added to Results, Neurological function Line 37, Page 13:*<br>"The RCT[30] showed an outpatient rehabilitation intervention had no significant effectiveness in improving cognitive function on performance-based cognitive tests compared to standard care at any follow-up point."<br><br>*Text added to Results, Neurological function meta-analysis Line 2, Page 14:*<br>"Five observational studies[20 26 27 32 33] were included in a meta-analysis. This showed rehabilitation significantly increased clinician-reported function…" |
| Have also performance-based tests been used in the relevant studies? | We have stated in the text where performance based tests were used. | *Text added to Results, Neurological function Line 37, Page 13:* |

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| | | "The RCT[30] showed an outpatient rehabilitation intervention had no significant effectiveness in improving cognitive function on performance-based cognitive tests compared to standard care at any follow-up point." |
| Have observer-reports consisted of clinician-reports (as we suspect is the case with FIM) only, or also of informant-reports involving next of kin or a close friend? If so, please state that this has been the case. | Clinician-reported function was used as an outcome in five studies. This information has been added to the results section.<br>No study reported informant-reports involving next of kin or a close friend, therefore no comment has been made on this in the text. | *Text added to Results, Neurological function meta-analysis Line 2, Page 14:*<br>"This showed rehabilitation significantly increased clinician-reported function," |
| We also know from studies on other acquired brain injuries or neurodegenerative disorders that the overlap between subjective (patient-reported) and objective outcome is not 100%. | We agree that outcomes measures may produce different results depending on whether they are observer or patient reported. In our meta-analysis we did not include outcome measures from different sources and therefore this did not influence the review's results. Again, given the limited space, we have decided not to include this point in the discussion but have instead focused on some of the other methodological considerations that may have influenced the review's results. Let us know if the reviewer thinks otherwise. | |

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| Furthermore, cut-points and instruments used to describe e.g. cognitive impairment varies between studies. | None of our included studies reported positive results based on cut-points, and considering the word count restrictions we have chosen not to elaborate further on this subject this within the current paper. | |
| The usage of pooled SF-36 data has been discussed as a limitation for the HRQoL data, but the authors may similarly want to discuss the implications of this as well. | The transformation of SF-36 scores from sub-scales to component scores in two studies allowed results to be pooled but, as you say, we have noted the potential limitation of some overlap in sub-scales. Determining the possible implications of this would be complex and a true and complete answer is not possible within the word limitations of this paper. Given this and that any implications are unlikely to influence the overall findings of the review, we have chosen to explore some of the other methodological implications that you have suggested and not this point in this paper. If the reviewer thinks differently please let us know. | |
| - Suggestion: Include more description of what we know about rehabilitation interventions in associated groups of patients (that we know often include a few CA survivors) e.g. brain injury rehabilitation, cognitive rehabilitation, psychological interventions, cardiac rehabilitation etc. to show that there are available interventions that could be tested for this group specifically as well. | This is an important consideration in the wider context of CA-survivor rehabilitation. We have addressed the limitations of looking across groups of patients but to go more in depth is to us outside of the scope of this review. | |
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| Minor comments | | |
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| Abstract | | |
| - Secondary consequences in the aim is a bit unclear, please specify what you mean with this. | Clarification on the specific secondary consequences has been added. | *Text added to the Abstract, aim Line 31, Page 1:* |

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| | | "physical, neurological and psychological" |
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| - We are uncertain of the dash usage in "8-weeks" and "2-years". | The dashes have been removed. | *Text corrected in Abstract Line 6, Page2*:<br>"8 weeks and 2 years" |
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| - The SMD of physical but not mental HRQoL is reported in the abstract. | This has been corrected. | *Text corrected in Abstract Page 2, Line 7:*<br><br>"…mental HRQoL (SMD 0.27 (95% CI: -0.01 to 0.55)." |
| Introduction | | |
| - SCA is primarily used to describe sudden cardiac arrest, even in scientific literature. This abbreviation appears throughout the manuscript and is somewhat confusing. Have you considered another abbreviation for survivors of cardiac arrest (such as CA-survivors)? | This has been changed throughout the manuscript. | *Text revised throughout manuscript:*<br>"SCA" changed to "CA-Survivors". |
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| - Secondary physical, neurological and psychological consequences – physical may also be of neurological reason. Do you actually mean physical, cognitive and psychological? | We agree the chosen terms could be considered to interact and there is perhaps no perfect solution to describing the secondary consequences in three discrete terms. Indeed, the framing and terminology to describe secondary consequences of cardiac arrest differ across the literature.<br><br>To be consistent with the terminology in the protocol, which was based upon the core domains described by COSCA, we would like to keep the term 'neurological'. We have been explicit in the Methods that the outcome neurological function includes cognition and have added text to the neurological function results section to highlight where results relate to cognition. | *Text added to Results, neurological function Line 37, Page 13:*<br>"The RCT[31] showed an outpatient rehabilitation intervention had no significant effectiveness in improving cognitive function on performance-based cognitive tests compared to standard care at any follow-up point."<br><br>*Text added to Results, neurological function data not included in meta-analysis Line 16, Page 14:*<br>"Cognition, specifically executive function, is the primary outcome in one included ongoing trial,[36] with results due in 2024." |
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| - Page 3 row 11-13. We are not sure about the wording here ("in people"). | We have deleted the phrase ‘in people’ | *Text deleted in Introduction Line 8, Page 3 :*<br>“in people” |
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| Methods | | |
| - Summary measures. A suggestion is that the authors here or somewhere else briefly specify the "acceptable" range for SMD and I2, for instance. This may be helpful for readers not familiar with these statistical methods. | Information on interpretation of SMD and $I^2$ has been added to the Methods section | *Text added to Methods, Synthesis of results Line 36, Page 5:*<br>“If studies were considered clinically comparable, data was pooled using a random effects meta-analysis. SMD was calculated where the same outcome was reported but using different measurement tools with values of 0.2, 0.5 and 0.8 interpreted as small, medium and large effect sizes respectively. Separate analyses were conducted for RCT and observational studies. Study heterogeneity was examined using the Cochran Q test and quantified with $I^2$ statistic (statistical heterogeneity indicated by chi-square test p value <0.10 and an $I^2$ statistic >50%). All analyses were conducted using STATA 16 (version 16, StataCrop, TX, USA) statistical software. |
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| Results | | |
| - We would not consider FIM as a measure of neurological function. FIM is related mostly to activity even if it includes some items regarding neurological or cognitive impairment. Problems may however not only be related to neurological function. | We agree that FIM may be considered a measure of activity and could be categorized differently. But as the FIM also includes two sub-scales on cognition and mobility and to remain consistent with the systematic review protocol, we have chosen to continue to categorize the outcome under neurological function.<br>To clarify this point we have added increased detail to the definition of the term neurological function as it is defined in the protocol. | *Text added to Methods, last paragraph Eligibility criteria Line 12, Page 4:*<br>“The main secondary outcome was neurological function, defined as measuring the level of disability after a neurological event. Measures may primarily test cognitive ability or may combine cognitive and physical ability hence measuring global disability.” |
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| - We are not sure if p=0.00 should rather be written out as p<0.001 throughout this section. | This has been corrected | |
| Also, the heading "HRQoL meta-analysis" should be in italics as well. | This has been corrected. | |
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| - Please be consistent regarding spacing before/after +/- signs. | +/- symbol has been replaced with SD as per the request of another reviewer | *Text amended throughout manuscript:*<br>SD replaces ± |
| Double check the number of parentheses used (e.g. in section "Synthesis of results, HRQoL studies not included in meta-analysis"). | This has been corrected. | |
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| Discussion | | |
| - The discussion of the different populations in the studies could be expanded. Some studies have only included CA related to cardiac cause, other studies have included patients with anoxic BI due to other reasons as e.g., suicide attempts or drug overdoses. This may influence on the results as well as the mix of IHCA and OHCA. | We agree that the mixed populations in two of the studies may have an effect on the results. A rough comparison of the FIM results suggests one of the mixed-population studies is not too different from the two others reporting FIM. However, without exploring the etiology and symptom trajectory of the different non-CA-survivors acquired brain injury patients, it is not possible to speculate further in what direction the results may have been influenced. Text has been added to explain this point in the Discussion.<br><br>Thank you for highlighting the potential influence of whether a study has mixed IHCA/OHCA or just IHCA/OHCA. We agree that this would influence the results. Unfortunately, there is not enough information on the proportion of IHCA/OHCA in the included studies in this systematic review to draw any conclusions about the influence of this on the effect of the stated rehabilitation interventions. Due to this, and the limits of space within the Discussion we have not addressed this point in the current manuscript but will | *Text added to Discussion Line 34, Page 17:*<br>"The effect of including studies with mixed populations on this review's results is difficult to determine without greater examination of the etiology and secondary consequences of the other non-CA causes of anoxic brain injury. However, Schmidt, 1997[32] showed a similar change in FIM to two of the studies[20 33] that only included CA-survivors (Fig. 6). |

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| | certainly consider it for future publications in this area. | |
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| - We do not follow the current statement "from these only associated change in outcomes can be determined not effectiveness" (page 15 row 9-10). | The phrasing has been changed to improve clarity. | *Text amended in Discussion Line 3, Page 16 :*<br>"Overall, quality of the body of evidence of these interventions is low or very low. Eleven of the fourteen studies were observational and cannot determine the cause and effect of an intervention, but can only show the associated change in outcomes between one time point and another" |
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| - We would classify stroke as an acquired brain injury (page 15 row 50). Consider rephrasing. | The phrase 'acquired brain injury' has been removed. | |
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| Table 1 | | |
| - Consider writing out the explanation for ICD as a footnote to the table. | This has been added. | |
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| Overall | | |
| - Be consistent in the use of primary outcome, main secondary outcome or secondary outcome. Define what is the primary or secondary outcome and describe this similar all over the manuscript and be clear in conclusions what outcome you refer to (e.g., if it is HRQoL or other). | The terminology related to outcome definitions has been corrected throughout. The outcomes have been more clearly stated in the conclusion. | *Text added to the Abstract, Conclusion Line 30, Page 1 and manuscript Conclusion Line 40, Page 18 :*<br>"Given the overall low quality of evidence, this review cannot determine the effectiveness of rehabilitation interventions for survivors of cardiac arrest on HRQoL, neurological function or other included outcomes, and recommend further high-quality studies be conducted." |
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| Reviewer: 2 | | |
| Dr. Frank H. Bosch, Hospital Rijnstate Arnhem | | |
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| Comments to the Author: | | |
| This study is well done research. The results are not surprising, but it is good that it is now established that there is not much literature on this topic; there are a few minor remarks regarding the text. | | |
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| Page 3: i think the authors meant (≥ 18 years) | This has been corrected | |

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| Page 5: continues data (i am not sure what they mean) | This has been corrected | |
| Page 6: GRADE was used; additional analyses: will be investigated. | The heading in the methods section has been amended to match the results section. | *Text amended in Methods Line 10, Page 6:*<br>"Risk of bias and quality of evidence across studies." |
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| Reviewer: 3 | | |
| Dr. Alex Prescuitti, University of Colorado | | |
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| Comments to the Author: | | |
| Thank you for inviting me to review this systematic review and meta-analysis on the effectiveness of rehabilitation interventions on secondary outcomes post-cardiac arrest. This is a timely and very important research question, given the call for multi-disciplinary rehabilitation efforts for cardiac arrest survivors. The authors found low quality evidence from 3 RCTs and 11 observational studies, and thus call for high quality RCTs comparing rehabilitation interventions with standard care alone. The study authors should be credited for utilizing an extensive search strategy and for utilizing appropriate tools to ensure the integrity of their appraisal of individual studies and subsequent analyses. | | |
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| Overall, the message is clear – high quality RCTs are needed to evaluate the effectiveness of rehabilitation interventions, and particularly, specific components of these interventions, in cardiac arrest survivors. | | |
| I only have a few points for consideration: | | |
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| Results: | | |
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| 1. Dougherty 2004,2005 does not appear to be a rehab intervention but rather an information packet aimed at increasing self-efficacy and reduce anxiety. I would consider excluding it. | We agree that the definition of rehabilitation we have used in the manuscript needs further clarification. No unique definition of rehabilitation can be found in the literature. This typically changes in relation | *Text added under Methods, eligibility criteria Line 4, Page 4:*<br>"Rehabilitation can be defined as: 'A set of measures that assist individuals, who experience or are likely to experience disability, to achieve and maintain optimum |

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| | to profession, health setting and patient populations. In designing the review protocol we used the White book definition:<br>'A set of measures that assist individuals, who experience or are likely to experience disability, to achieve and maintain optimum functioning in interaction with their environments.<br>(From: Cantista P, Ceravolo MG, Christodoulou N, et al. White Book on Physical and Rehabilitation Medicine (PRM) in Europe. Chapter 1. Definitions and concepts of PRM. *European Journal of Physical and Rehabilitation Medicine* 2018;54(2):156-65. doi: 10.23736/S1973-9087.18.05144-4).<br><br>As this is the first review of rehabilitation interventions for CA-survivors we wanted to include all possible rehabilitation interventions. We therefore used the White book definition since as it gives a broad difination of rehabilation in line with the broad needs of CA-survivors and the aim of this review. Hence, the PICO 'intervention' eligibility criteria were defined not be inclusion of specific interventions such as physiotherapy or cardiac rehabilitation but, in alignment with the White book definition of rehabilitation, by the exclusion of primarily pharmacological, surgical or invasive technology.<br><br>The White book definition of rehabilitation we have used has been added to the text and we have added detail to the search matrix terms we used. | functioning in interaction with their environments'.[7] To align with this broad definition of rehabilitation and ensure inclusion of all possible rehabilitation interventions, interventions were included if they were not primarily pharmacologically or surgically-based or involve invasive technology. Interventions in the emergency room or critical care unit setting were excluded."<br><br>*Text added under Methods, searches Line 39, Page 4:*<br>"The search matrix consisted of a combination of keywords and synonyms for: 1) cardiac arrest, and 2) non-pharmacological/surgical/invasive technology rehabilitation interventions"<br><br>*Text added to Table 1:*<br>1) Booklet mailed to study participants on strategies to manage recovery, 2) structured information provided by experienced cardiovascular nurses to improve self-efficacy to deal with illness demands, and to reduce anxiety, through identification of illness related problems and behavioral strategies to manage them including role playing, problem-solving, goal-setting and collaborating on the learning assignment for the coming week. |

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| | The full text of the study articles were re-screened by the two authors who undertook the original article screening. The study was deemed to meet the review inclusion criteria. However, we acknowledge that more detail on our definition of the PICO 'interventions' and the interventions in Dougherty, 2004, 2005 should be provided to support the decision to include this study.<br><br>Based on our explanation we hope the reviewer supports that we leaves the study in our review. Otherwise let us know. | |
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| 2. It is unclear if Dougherty 2019 was a rehab intervention based on limited detail provided. | We agree that the definition of rehabilitation we have used in the manuscript needs further clarification. Please see comment above for our full explanation.<br><br>The full text of the study articles were re-screened by the two authors who undertook the original article screening. The study was deemed to meet the review inclusion criteria. However, we acknowledge that more detail on our definition of the PICO 'interventions' and the interventions in Dougherty, 2019 should be provided to support the decision to include this study.<br><br>Based on our explanation we hope the reviewer supports that we leave the study in our review. Otherwise let us know. | *Text added as in the previous comment above.*<br><br>*In addition:*<br>*Text added to Table 1:*<br>2) Nurse telephone support to improve self-efficacy and problem solve. Components of this support were as per the intervention in Dougherty, 2004, 2005. |
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| Minor: | | |
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| Methods: | | |
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| 1. Edit to ensure tense remains consistent in the Methods | This has been corrected in the Methods section | |

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| section as the authors used both past and present throughout. | | |
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| 2. At points the primary aim of the study is reported to be examining effectiveness of rehab interventions, with safety being a secondary outcome. At other points the primary aim is reported to be examining effectiveness and safety. Please edit for consistency. | This has been corrected throughout the manuscript. | |
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| Reviewer: 4 | | |
| Dr. Adrian Esterman, University of South Australia | | |
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| Comments to the Author: | | |
| Page 4: Introduction, line 47 - Should read ≥18 | This has been corrected in the Introduction | |
| Page 6: Summary measures, line 35- Should read continuous data | This has been corrected in Summary measures | |
| Page 6: Synthesis of results, line 47 – Should read Study heterogeneity, not Study variance | This has been corrected | |
| Page 13: Synthesis of results, lines 31-35 – please use p<0.001 for values close to zero. | This has been corrected | |
| Page 13: Line 41, please do not use ±. If these are standard deviations, say so. This is true throughout the results. | The text has been amended to replace ± with SD | *Text amended throughout manuscript:*<br>± symbol has been replaced with SD. |
| Pages 29 and 30: Both forest plots labelled Figure 4B! | This has been corrected. | |
| In the first, I2 was 0, yet a random effects model was still used. | We followed the Cochrane Handbook guidance and made the choice between a fixed-effect model and a random-effects meta-analysis before data analysis and not on the basis of the statistical test for heterogeneity (Handbook 10.10.4.1). The choice of a random effects model was made because there was some clinical heterogeneity between the studies, with some small differences in intervention, outcomes and study population. Hence, we could not use a fixed model as the underlying assumption would be that all variation in study effect | |

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| | is only random variation, in random effects models it is reserved that it cannot be ruled out that there may be differences in intervention effects (due to the clinical heterogeneity).<br><br>Based on our explanation we hope the reviewer supports our choice of using a random effects model. Otherwise let us know. | |
| Also for Table 5, I2 was 17%, yet a random effects model was still used | We followed the Cochrane Handbook guidance and made the choice between a fixed-effect model and a random-effects meta-analysis before data analysis and not on the basis of the statistical test for heterogeneity (Handbook 10.10.4.1). The choice of a random effects model was made because there was some clinical heterogeneity between the studies, with some small differences in intervention, and study population. Hence, we could not use a fixed model as the underlying assumption would be that all variation in study effect is only random variation, in random effects models it is reserved that it cannot be ruled out that there may be differences in intervention effects (due to the clinical heterogeneity).<br><br>Based on our explanation we hope the reviewer supports our choice of using a random effects model. Otherwise let us know. | |

**VERSION 2 – REVIEW**

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| **REVIEWER** | Lilja, Gisela<br>Lund University, Department of Clinical Sciences |
| **REVIEW RETURNED** | 04-Jun-2021 |

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| **GENERAL COMMENTS** | I did this review toghether with psychologist Erik Blennow Nordström, PhD student. We both think that the authors responded well to our comments. Especially we are happy to see the definition of rehabilitation used, this helped us to understand their approach. In addition the discussions now more includes clearly includes the limitations of the study. We have no further questions and wishes the authors good luck with this important paper. |

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| **REVIEWER** | Prescuitti, Alex<br>University of Colorado, Psychology |
| **REVIEW RETURNED** | 23-May-2021 |

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| **GENERAL COMMENTS** | Thank you for addressing all comments and making appropriate changes. |